

# *Image Analysis and Detection of Tea Leaf Disease using Deep Learning*

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**Abstract**— The widespread development of deep learning has directed to the increasing research interest in image recognition technologies, which enables in the field of computerized image categorization and finding of plant diseases. India is the world's largest tea consumer and the second largest tea producer. Diseases can harm the plant's health, which in turn affects its growth. There are a number of tea leaf diseases that target the rate at which the crop grows. The purpose of this paper is to develop a deep CNNs called LeNet to discover the tea plant diseases from leaf image set. As a result, LeNet the perfect CNN model used in further applications to improve the diagnostic measurement of tea leaves as well as other plant leaves.

**Keywords**— *Convolutional Neural Network (CNN), LeNet.*

## I. INTRODUCTION

India produces the largest amount of tea, due to land availability. Indian Tea Giant Goodrich Group Parent Camellia PLC will be the largest private tea producer in the world by 2018, producing 103 million kg. Tea production in India was virtually non-existent until the end of the 16<sup>th</sup> century [1]. The British people made a series of attempts to improve the tea cultivation rates in India. The results of their cultivation work in Darjeeling and Assam (the largest growing areas of Indian tea) have exceeded their expectations by putting a lot of efforts [2-3].

High yields of tea are mainly achieved by removing shade trees and reducing the losses caused by pesticides and pests. The main cause of blister blight is greater in the economic loss of tea compared to animal pests [4-6]. The current method of diagnosing plant diseases is simply a manual inspection from their own eyes, which requires much man power and laboratories equipped with expensive equipment's [7]. And improper illness detection with inexperienced people and lot of chemical consumption which will cause the growth of long life struggle of the pathogen and dropping the immunity of the tea plant to fight back [8].

Tea plant diagnoses can quickly increase the complexity of the plant's leaves by naked eye care. The

large amount of cultivated crops and its diagnostics problem can lead to an experienced agronomist and botanists may fail to successfully identifying the specific type of disease. It can also lead to the false predictions and conclusions [9]. An autonomous system may help to discover plant syndromes by the color and visual indications of the plant may greatly assist even non-technical persons could able to tackle the problem in the agricultural system. This will prove to be an effective practice for agriculturist, and put aware of them at the earliest before the outbreak of large-scale disease [10].

At this period of time, deep learning systems have been shown to surpass previous sophisticated machine learning techniques in many fields particularly in computer vision one of the most important cases [11]. The first and important reason for deep learning success is to automatically integrate feature extraction process into the training sector. Deep learning has succeeded in collection of large data sets and systematically integrates performance appraisal into the development process [12]. The objective of this proposed work is to enable automated detection of diseases in tea leaf with efficient computer-aided neural network model. The method used here to spot plant diseases is image processing using the convolution neural network (CNN), the LeNet model [13]. I hope that if our program is ever implemented in real life it will help farmers to avoid their losses. Using the CNN LeNet algorithm, the disease-induced leaf layer classification has been performed.

## II. TEA LEAF DISEASE

The data leaves in the Fig.1 contain 80 images of selective tea leaf's diseases, named as blister blight, red scab, red leaf spot and leaf blight. The image is collected from plant village dataset.

### A. Illness indications:

Little, pinhole-size spots territory unit from the start seen on youthful leaves. On the off chance that, the spots in the leaf becomes clear, bigger, and lightweight darker. Subsequent to observing seven days, the lower leaf surface creates rangle like side effects [14].

### B. Tea Red Leaf Spot disease

This is known as the Boria root disease of tea. The first symptoms appear as a yellowing of the leaf, and then wither, and then they are suddenly attached to the stem for several days after death with bushy or full bushy atmospheric leaves [15].

### C. Tea Red Scab

The high isolation rate of *Phyllosticta* sp. because of its ascent and tidy agamogenesis might have crystal rectifier to the incorrect identification of this flora because the reason behind tea white scab sickness.

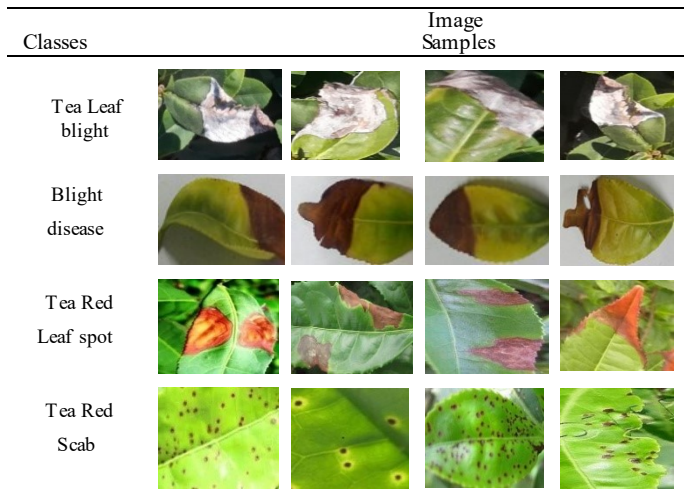


Fig.1. Tea Leaf Image Data Samples

## III. RELATED WORKS

In the following cases, some early works undergone in this field are reviewed for reference.

The different classification techniques used to diagnose plant species. For a given test example, the K-nearest-neighbor clustering method uses class prediction. To do that, the image is turned into a gray scale. If the given training data is not linearly separated, then the support vector machine is not converged well, which appears to be one of its drawbacks [1].

Prof. Sanjay B. et al., [2] describe that there square measure solely follows 3 steps in developed model out of this, the primary one is, for the given input RGB image, a color transformation structure is made, that is, HSI is employed for color descriptor. In second step, by selecting the brink price from bar chart, inexperienced pixels square measure disguised and removed and also the Region of Interest that square measure extracted. And in last or third main step the classification is completed.

The authors of this study [3] present a technique for classifying and identifying different diseases affecting

plants. The technique presented in this paper for color package extraction is the color co-occurrence method. Artificial Neural networks were used to detect diseases in the leaves automatically. The proposed approach may substantially support the accurate diagnosis of the leaf, and appears to be an important approach in the case of vapor and root disease, but with less effort in computation.

In this paper the disease identification process follows the four main steps: first the given input image in RGB color format is changed to HSI color format. Then the green pixels in the image are marked and isolated using a threshold value. In the third step the segmentation process is followed, to obtain the region of interest from the image. Finally, the classifier is used for the diagnosis the disease based on the features. The robustness of the proposed method has been demonstrated on the 500 plant leaves from the data set [4].

According to paper [5] histogram matching technique is employed to spot disease. Histogram matching, edge detection methodology and extracting color features play an important role in this paper. From the plotted histogram the layers of RGB image is divided into red, green, and blue layers and then applying the edge detection algorithm in retrieved image.

This paper [6] presents methods of image analysis through edge-based thresholding; K-means clustering and region-based extraction. The extracted regions of objects were compared to the mean difference value between the object. The correlation of objects is calculated in two images, to measure the similarity index between the images. The PCA algorithm is used to reduce the quantity of the feature with different orientations to improve the classification accuracy. This approach tells us that the K-mean clustering approach is good in yielding high accuracy compare with other approaches.

According to this study [10], fruit detection using color and texture to separate region of objects in the image was evolved. The author presents a model for diagnosing and classifying diseases using the k-means clustering approach. Multiple samples are taken to apply in the segmentation algorithm to classify fruit and disease. First, image acquisition is done to capture the image and remove the effects of the hardware used for fabrication in the preprocessing stage, and the image is fragmented using segmentation algorithm. Finally, the feature extraction is performed and then fitted with the training dataset for image classification.

All the above-named papers are totally studied. These papers gave tons of knowledge regarding the chosen topic, the assorted counseled algorithms. And therefore the basic foundation construct tends to be discovered from the top of the mentioned papers. And therefore the recent advancements in deep learning were revised.

#### IV. PROPOSED METHODOLOGY

The overall model of this work is represented in the following Fig. 2.

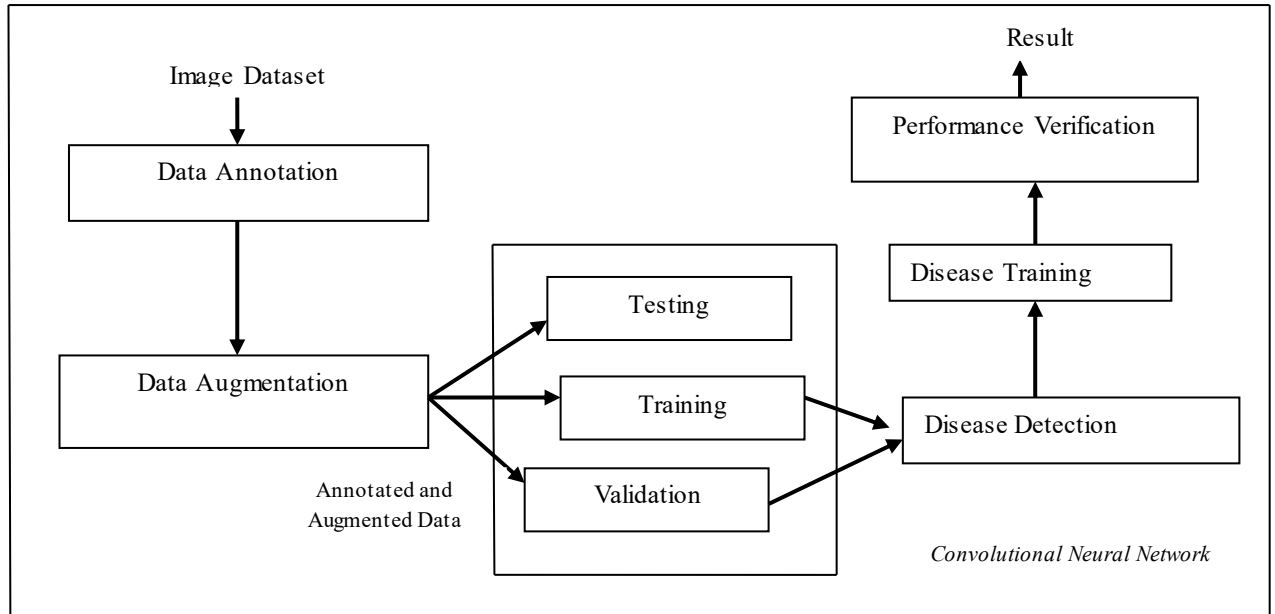


Fig.2. The Proposed model

##### A. Image Annotation And Augmentation

Image annotation [8], is the process of making explanatory terminology for an image by automatically, which plays an important visual clue for the images in the searching and retrieval applications. In this model, the various parts of the images with the disease are marked using border box technique. Thus the annotation process can be used to name the image and location of the affected area in the leaf. Fig.3 shows the annotated image.



Fig.3. Annotated Image

The various formats of images are collected with different resolutions and orientation. In order to obtain the best accuracy in the extracting feature, the images must be correlated with the deep neural network, which were pre-

processed to obtain uniformity by automatically. Images used for this application were changed to  $256 \times 256$  to reduce the training time period.

There is one of many transformation techniques involved in image development including affine transformations, perspective change, image rotations and intensity changes. This technique is simply called augmentation [11] and it is played a crucial role in the deep learning preprocessing stage. Fig. 4 and Fig. 5 show an example of affine transitions and simple rotations.

##### B. Convolutional NN Overview

The basic ideology of CNNs [7] is commenced by LeCun et al. within the late 1990s. The deep neural networks are comprised with n number of layers with extraneous features. The best deep neural network identified for our proposed work is Convolutional Neural Network (LeNet model). Three differing kinds of layers (Convolution, pooling and fully connected layer) are distinguished during a CNN operation. The first layer is convolution which is the essential foundation of this network. This part of layer computes a convolved input using a filter simply a mask value in a linear manner. The coefficient value of this kernel is similar to the weights in the neural network. The coefficient values are updated as similar to weight in the conventional neural network. Convolution is simply a pixel wise matrix multiplication

along with the filter values. Thus the main aim of extracting the high level features from the input image was made successfully. The output of this layer was convolved feature matrix and the filter used for convolution is known as kernel.



Fig. 4. Affine transformations



Fig. 5. Rotations

Fig.4, 5. Augmented Image

Then the pooling layer is followed by convolution layer. The main purpose of pooling layer is to down sample the images. It is used to progressively reduce the size of an image. Pooling plots calculate the average value over a portion of the feature map. There are two types of pooling operations are there such as max pooling and average pooling. And the max pooling is the most preferable one. It also performs de-noising along with the dimensionality reduction.

Convolutional layers and pooling layers both are playing an important role in to localize the high level features from the image. The convolutional neural network produces a good performance in computer vision with added feature of equipped with many number of layers and neurons that have not seen in the traditional network. In

contrast, if static nonlinear neural networks are used for processing the raw (or slightly pre-processed) images, the images are vectorized in the first step. The result was act as inputs of the neural network. By doing this process, information about the spatial relationship between the pixels was discarded, and the feature extraction process was done by an efficient segmentation algorithm which is not seen in convolutional neural network. But in the proposed model, the feature extraction process is done by automatically.

The final layer in the CNN is responsible for the classification work which is typically a multi-layer neural network. It receives dimensionality reduced feature maps as input from the preceding layer. The ultimate layer is that the output layer which is the classification layer. The softmax activation function is used for tuning the input into a likelihood sharing over the classification. The softmax activation function gives the highest classification accuracy.

In this work, a classic convolution network model called as LeNet-5 is chosen. This LeNet-5 model has 2 convolution layers followed by 2 average pooling layers which is also called sub-sampling and then followed by single flattening convolution layer then two fully connected layer with one softmax classification layer. The over read of Convolutional Neural Network model is portrayed within the following Fig.6. From the figure, the given 3 X 227 X 227 input image is convolved with 3 X 11 X 11 kernel filter to obtain the image dimension with 24 X 55 X 55 pixels. This is applied to avg pooling layer to reduce the image dimension and the result is 64 X 27 X 27 pixels. Further proceed with convolution and pooling layer to fine tune the network. The 500 feature maps with the size 1 X 1 in the fully connected layer are mapped into 10 feature maps in the next connected layer. The last softmax classification has the 4 values with 4 different diseases of tea plants.

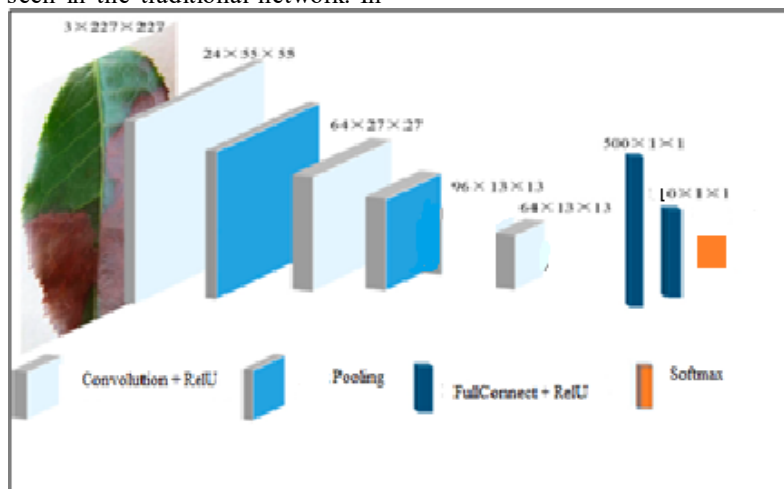


Fig.6. The structure of convolutional neural network model

## V. RESULTS AND DISCUSSIONS

The accuracy of CNN classifiers was evaluated using ROC(Receiver Operative Curve) in diagnosing the disease status for various tea leaves from the image set. The outcomes of this evaluation are shown in Fig. 7. Sensitivity and specificity were used to assess the accuracy of the tea leaf disease diagnosis system (Table 1). In different diseases, Blight leaf disease is producing highest accuracy level due to the apparent phytopathological symptoms and the simplicity of prudence. Tea red scab is the next most accurately classified disease with yield 94%, and all other remaining diseases are classified as 84-93%. From the identified accuracy levels, it has been believed that this model is producing a high accuracy level compare with traditional backpropagation network which yield 85% accuracy for the blister blight disease.

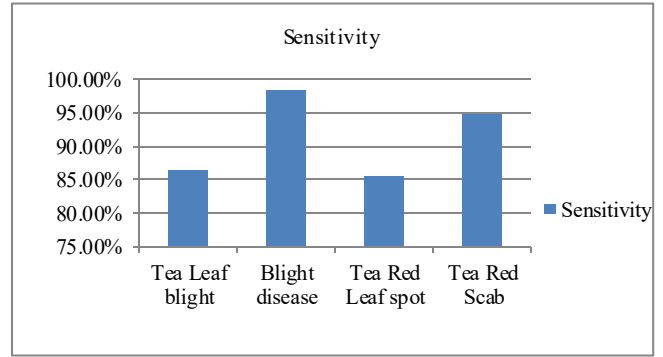


Fig.7. Accuracy (as a %) of disease classification

TABLE I. Error matrix showing the classification accuracy of the LeNet algorithm

|                   | Tea Leaf blight | Blight disease | Tea Red Leaf spot | Tea Red Scab | Sensitivity | Accuracy | MCA    |
|-------------------|-----------------|----------------|-------------------|--------------|-------------|----------|--------|
| Tea Leaf blight   | 90              | 1              | 0                 | 0            | 86.49%      |          |        |
| Blight disease    | 1               | 117            | 0                 | 0            | 98.32%      | 90.23%   | 90.16% |
| Tea Red Leaf spot | 0               | 0              | 95                | 7            | 85.59%      |          |        |
| Tea Red Scab      | 0               | 3              | 0                 | 111          | 94.87%      |          |        |

## VI. CONCLUSIONS

The computational complexity required for using conventional neural network is significantly reduced using this model. The high adaptation to internal failure of CNNs permits the utilization of incomplete or uncertain foundation pictures, subsequently adequately improving the precision of picture acknowledgment. The consequences of the current examination underline the chance of utilizing CNNs in the recognizable proof and assessment of tea plant sicknesses, which could altogether improve malady acknowledgment for tea ranch horticulture. In spite of the fact that the exactness of LeNet's sickness characterization isn't 100%, upgrades to the present strategy might be actualized in future examinations. In our future work, the LeNet model has been utilized to distinguish other yield ailments, for example, tomato leaves and mulberry leaves. However, this should be further investigated according to the specific circumstances of the disease.

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